

1(a)(i)	region (of space)	B1
	where a particle experiences a force	B1
1(a)(ii)	force per unit mass	B1
1(b)	$g = GM / R^2$	C1
	$= (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (3.39 \times 10^6)^2$	C1
	$= 3.73 \text{ N kg}^{-1}$	A1

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Question	Answer	Marks
1(c)	$0.99 \times 3.73 = (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / r^2$	C1
	$r = 3.41 \times 10^6 \text{ (m)}$	C1
	height = $(r - R)$ $= 2 \times 10^4 \text{ m}$	A1
	or	
	$0.99 \times 3.73 = (6.67 \times 10^{-11} \times 6.42 \times 10^{23}) / (R + h)^2$ $(R + h)^2 = 1.1596 \times 10^{13}$	(C1)
	$R + h = 3.41 \times 10^6 \text{ (m)}$	(C1)
	$h = 2 \times 10^4 \text{ m}$	(A1)
	or	
	$0.99 = (3.39 \times 10^6)^2 / r^2$	(C1)
	$r = 3.41 \times 10^6 \text{ (m)}$	(C1)
	height = $2 \times 10^4 \text{ m}$	(A1)